Welcome.

Everyone:

- Pull the updates from the course GitHub repo:
  - `cd <46120-PiWE repo>`
  - `git pull origin main`

NB:

- By attending this class, you consent to being recorded. Recording will be shared to this class and possibly other DTU students for training purposes.

46120: Scientific Programming for Wind Energy

Final project introduction

Ju Feng

Agenda for today

- Pull new course material ✔
- Final project introduction
- Begin homework

Final project introduction

Important deadlines

- **Form group (2-3)**: by Sunday, March 29th, at 23:59
  - **Ask for custom project**: by Wednesday, April 8th, at 23:59 (email Ju)
- **Hand in final project**: by Monday, May 4th, at 23:59

Grading

Your individual final grade (pass/fail) will be determined as a cumulative evaluation of your contributions to the final project and an individual written quiz.

Selection of projects

We provide three pre-defined final projects

- **Wind resource assessment based on reanalysis data**
- **Wind turbine modeling based on Blade Element Momentum (BEM) theory**
- **Wind power forecasting using machine learning**

You may also submit a request for a **custom project** if your group have an idea you would like to pursue. Your custom project will need to fulfill a list of requirements to get approval.

Requirements to pass

1. Your final project should have the folder structure and required files as shown in the diagram:
2. Your Python package inside the `src` folder should include at least one class (we will cover class and object oriented programming in week 9).
3. Your Python package should implement the required functions, either as listed in functional requirements in the pre-defined projects, or as we agreed on and documented if you work on a custom project.

```
[your_final_project]
├── inputs/
│   ├── data_files_we_provide
│   └── data_files_you_found (optional)
├── outputs/
│   └── data_files_you_generate (no need to push to Github)
├── src/
│   └── your_python_package_codes
├── tests/
│   └── python_scripts_you_write_for_tests
├── examples/
│   ├── main.py (will run in evaluation)
│   └── other_example_scripts_you_write (optional)
├── .gitignore
├── LICENSE
├── Collaboration.md
├── README.md
├── pyproject.toml
└── any_other_files_you_may_want (optional)
```

Requirements to pass (continued)

4. The README file should contains:
   - A brief overview of the package objective.
   - Installation instructions.
   - A description of the package architecture, with at least one diagram.
   - A description of the class(es) you have implemented in your package, with clear reference to the file name of the code inside `src`.
   - A description of what peer review (if any) you have implemented.
5. We should be able to install your package successfully following the Installation instruction in your README.md.
6. Your `main.py` script inside the examples folder should demonstrate, in a clear and structured manner, how the required functions are called and executed.
7. Your `main.py` script should run successfully and generate the expected results (like plots) in a reasonable time (less than 10 mins). If needed, define a "demo" mode and/or use saved intermediate results.

Requirements to pass (continued)

8. Test coverage on your package should be higher than 80%, as evaluated using `pytest-cov` on your `src` folder, i.e., by running:

   ```
   pytest --cov=src tests/
   ```

9. pylint score on your package should be higher than 8.0, as evaluated using `pylint` on your `src` folder, i.e., by running:

   ```
   pylint src/
   ```

Questions?

Pre-defined projects

- Go to GitHub

Important notes

- The final project **is much more open** than the CodeCamp project, which essentially only lists the requirements, while **all the design and implementation details are left for you to decide**.
- This is not by accident, as **the only way to actually learn how to develop tools/packages to solve realistic engineering problems is to actually do it from scratch by yourselves**.
- PiWE focuses on scientific programming, not on the actual models or physics. **You don't need to be an expert** in wind resource assessment, BEM theory and machine learning/forecasting to do the final projects.
- Remember that **the evaluation of the final project focuses more on the structure and quality of the code, not on the performance of your developed model/tool**.

Ask me and TAs if you have any related questions on the models/physics.

A suggested development path

We have a tight schedule, think about your group's development plan. Below is a suggestion to consider:

- **Week 8 and 9**: Understand the project (background, requirements and necessary steps/models/equations), explore the provided data (load, parse, visualize), and consider the preliminary design of the architecture of your package.
- **Week 10**: Implement and refine different parts of the package (including tests).
- **Week 11**: Present your architecture and preliminary results in class, finalize the architecture and polish the code (including tests).
- **Week 12**: Refine the documentation, experiment on different computers (install the package, run the main script, check the test coverage, pylint score), fix all potential issues.

Check the relevant notes

Notes and Tips for this course

This folder contains a list of notes and tips for selected topics, which are important for the programming assignments/projects in this course.

Currently, we have:

1. Interpolation in Python: An Introduction with `NumPy` and `SciPy`
2. Introduction to Machine Learning with `scikit-learn`
3. Introduction to Curve Fitting with `SciPy`
4. Working with Wind Direction in Python
5. Organize a Python Package with `pyproject.toml` and `__init__.py`

Homework for this week.

Clean up your codecamp code.

Homework

- **PART 0: Sign up for a final project team (ASAP)**
- **PART 1: Explore one of the pre-defined projects**
  - Discuss with your group, choose one pre-defined project to explore.
  - Write a function to load, parse and plot the provided dataset in inputs, generate at least one figure. Ideally also with test in tests.
  - Think about the structure/architecture of your package.

  Note: the project you choose in this homework doesn't mean you have to stick to that project. Your group can make decision later on which project to work on. If you have a better project idea that your group want to pursue, first check the section on Custom project, and then come talk to us.

  But remember to make a decision early, so that your group has enough time to work on it.